



# 第八章

# 电容式传感器

*(capacitance sensor)*

# Capacitive Sensor

- Capacitive sensing (sometimes capacitance sensing) is a technology, **based on capacitive coupling**, that can detect and measure anything that is conductive or has a dielectric different from air.
- **Capacitive sensors use capacitive sensing**, including sensors to detect and measure proximity, pressure, position and displacement, force, humidity, fluid level, and acceleration.
- Capacitive sensors constructed from many different media, such as copper, indium tin oxide (ITO) and printed ink.

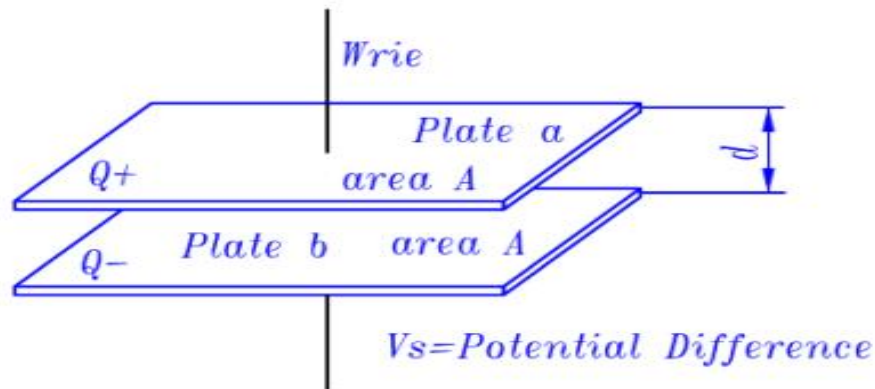
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- **3 Variable Permittivity Capacitive Sensors**
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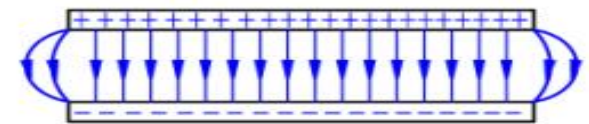
# 一、原理 Basic Principle

将被测物理量转换为电容量变化的装置，实质是一个具有可变参数的电容器。

The capacitance is defined as the ratio of the stored charge to the voltage applied



a. Arrangement of the capacitor plates

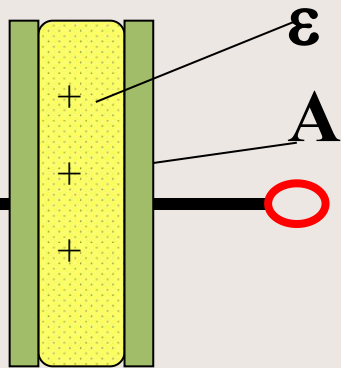


When separation of the plates is small compared to their size, the fringing of the electric field is slight

b. Side view of the electric field

# 8.1 Basic Principle

The capacitance value  $C$  depends on the plate size and dielectric material placed between the plates



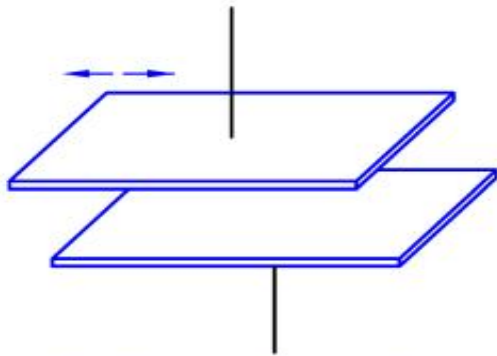
两个平行极板组成的电容器电容量：

$$C = \frac{\epsilon_0 \epsilon A}{\delta}$$

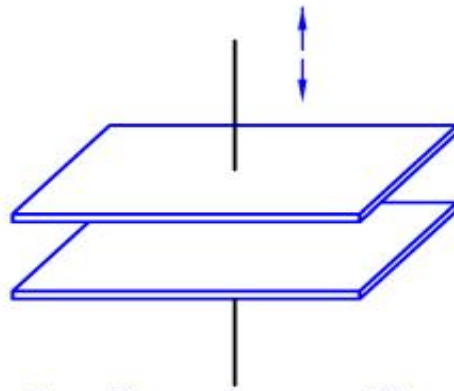
当被测量使  $\delta$ 、 $A$  或  $\epsilon$  发生变化，都会引起电容  $C$  的变化，若保持其中两个参数不变，而仅改变另一个参数，就可把被测参数的变化变换为电容量的变化。

据电容器变化的参数，可分为极距变化型、面积变化型和介质变化型三类。

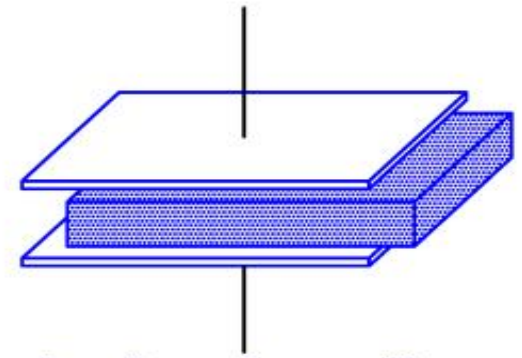
## Capacitance Change Patterns



*Capacitance changes with overlapped area change between plates*



*Capacitance changes with gap distance change between plates*

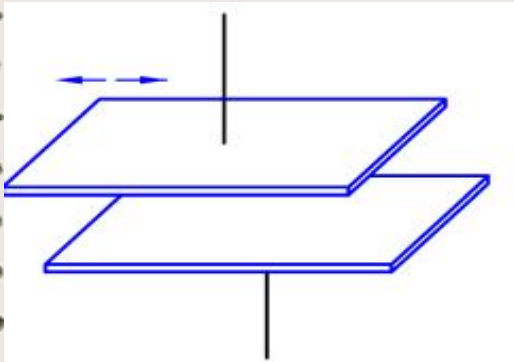


*Capacitance changes with dielectric change between plates*



## 8.2 Measuring Displacement and Velocity

### 8.2.1 Area changing type



$$C = \frac{\varepsilon A}{d} = \frac{\varepsilon W}{d} x, \quad \text{Farad}$$

the linear relation between the capacitance and displacement

$$K = \frac{\partial C}{\partial x} = \frac{\varepsilon W}{d}$$

## 8.2.1 Area changing type

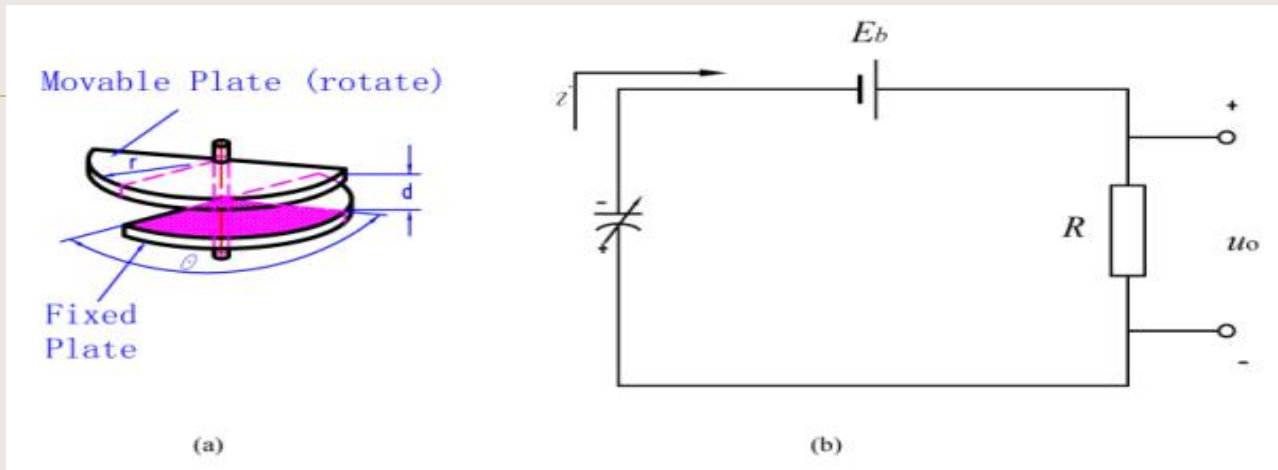


Fig 8.4 Capacitive Sensor used to Measure Angular Displacement

maximum value of capacitance

$$C = \frac{\epsilon A}{d} = \frac{\pi \epsilon r^2}{2d}$$

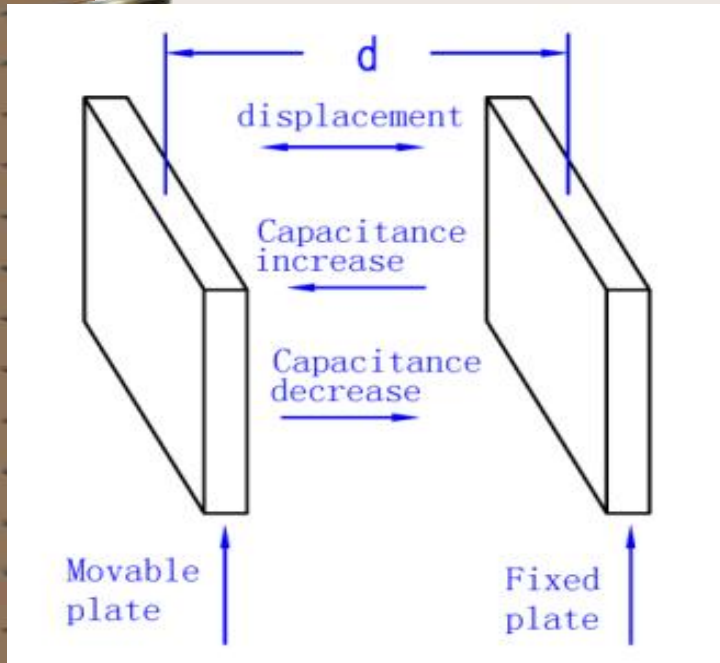
angle  $\theta$

$$C = \frac{\epsilon r^2}{2d} \theta$$

$$C = K \theta$$



## 8.2.2 Distance changing type



$$C = \frac{\epsilon_0 \epsilon_r A}{d}, \quad \text{in Farad}$$

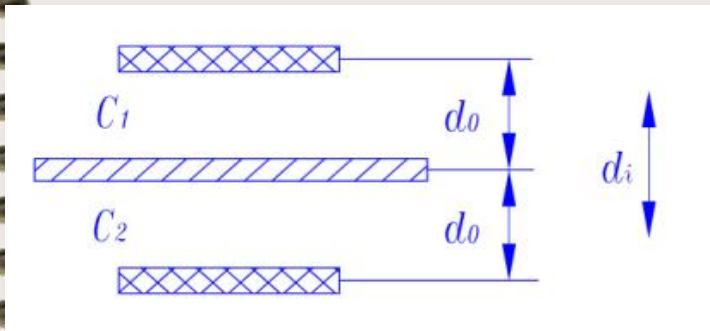
To get a linear output signal, a constant amplitude current signal  $I$  is now applied to the capacitor, making

$$V = \frac{I}{2\pi f C}$$

Where  $V$  is the voltage applied to the plates which is proportional to displacement  $\Delta d$ .

## 8.2.2 Distance changing type

### 1. Linearization by Taylor Series Expansion



The capacitance  $C_1$  and  $C_2$  can be written as

$$C_1 = \varepsilon_0 \varepsilon_r \frac{A}{d_0 - \Delta d} \quad (8.14)$$

$$C_2 = \varepsilon_0 \varepsilon_r \frac{A}{d_0 + \Delta d} \quad (8.15)$$

differential form of the capacitance

$$C_d = C_1 - C_2 = \varepsilon_0 \varepsilon_r \frac{A}{d_0 - \Delta d} - \varepsilon_0 \varepsilon_r \frac{A}{d_0 + \Delta d} = \frac{2\varepsilon_0 \varepsilon_r A \Delta d}{d_0^2 \left[ 1 - \left( \frac{\Delta d}{d_0} \right)^2 \right]}$$

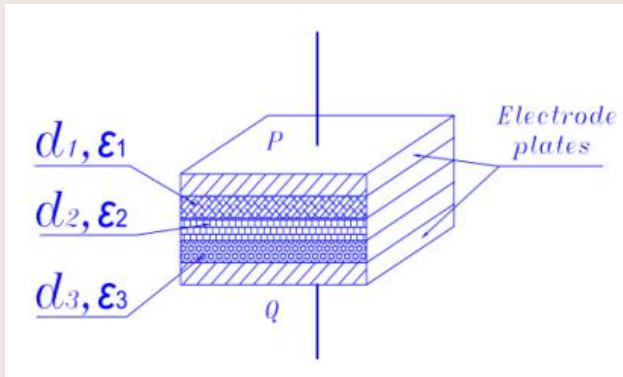
For  $\Delta d^2 \ll d_0^2$

$$C_d = \frac{2\varepsilon_0 \varepsilon_r A}{d_0} \frac{\Delta d}{d_0} \quad (8.17)$$

## 8.2.2 Distance changing type

if  $C_1$  is a free space air capacitor and  $C_2$  is a capacitor with dielectric  $\epsilon_r$ , whose permittivity is to be measured, the permittivity of the dielectric medium can be given

$$\epsilon_r = \frac{C_2}{C_1}$$

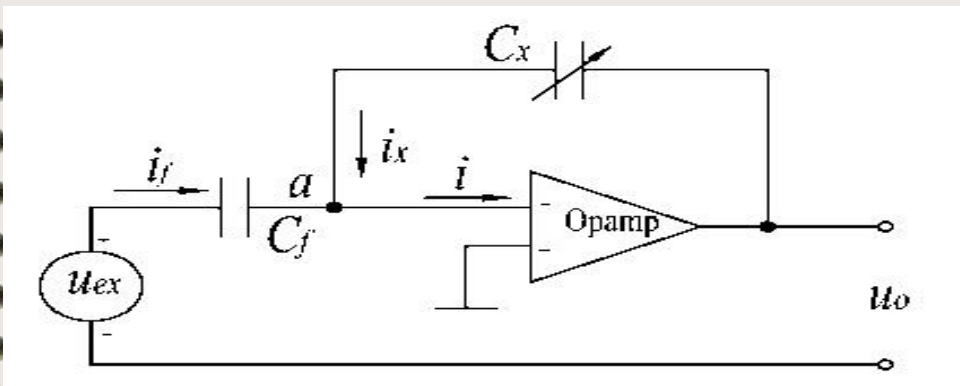


$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{A}{\frac{d_1}{\epsilon_1} + \frac{d_2}{\epsilon_2} + \frac{d_3}{\epsilon_3}}$$

Multiple dielectric layers

## 8.2.2 Distance changing type

### 2. Linearization Capacitive Displacement Measurement Using Operational Amplifier

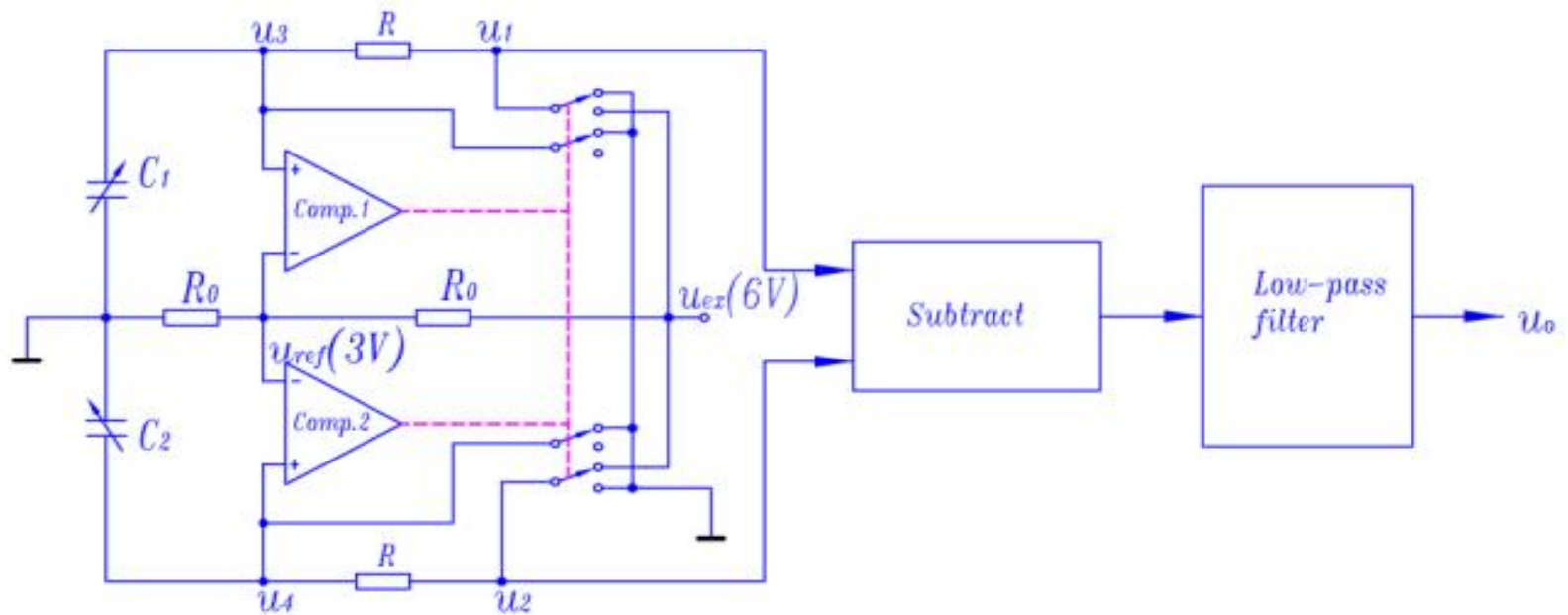


$$u_o = -\frac{C_f}{\varepsilon A} u_{ex} d = Kd$$

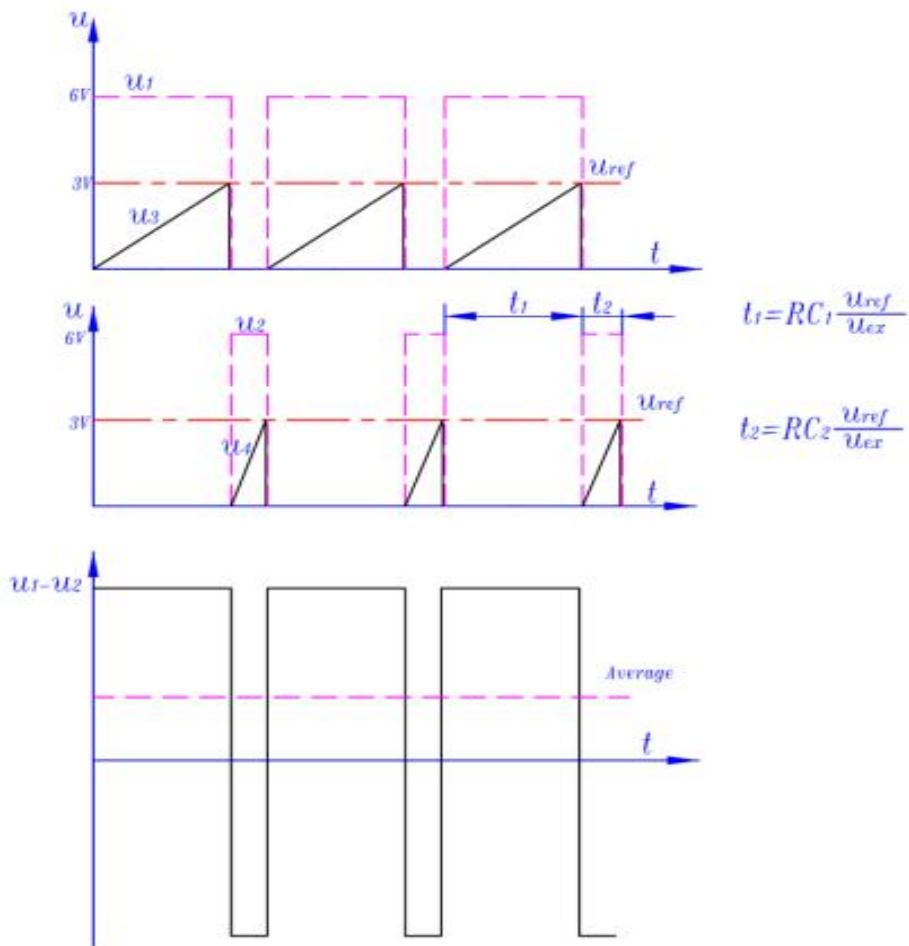
$$\text{where } K = -\frac{C_f}{\varepsilon A} u_{ex}$$

## 8.2.2 Distance changing type

### 3. Linearization Capacitive Displacement Measurement Using Pulse-Width-Modulation



## 8.2.2 Distance changing type



$$u_o = \frac{u_{ex} t_1}{t_1 + t_2} - \frac{u_{ex} t_2}{t_1 + t_2}$$

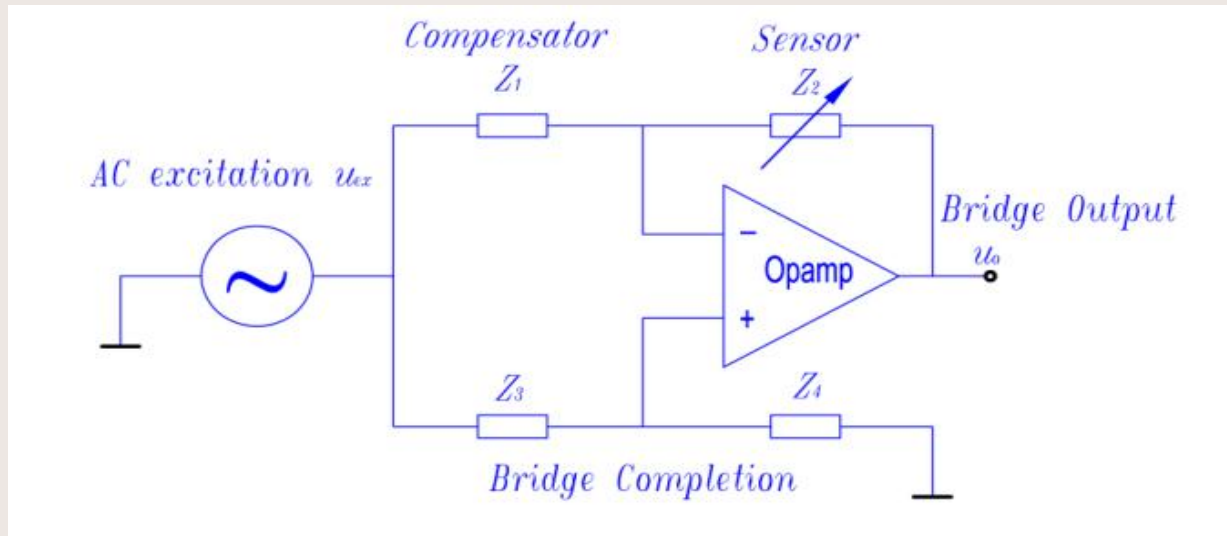
$$= u_{ex} \frac{C_1 - C_2}{C_1 + C_2}$$

$$C_1 = \frac{C_o d_o}{d_o - d_i} \quad C_2 = \frac{C_o d_o}{d_o + d_i}$$

$$u_o = u_{ex} \frac{d_i}{d_o}$$



## 8.2.3 Capacitance Bridge Circuit

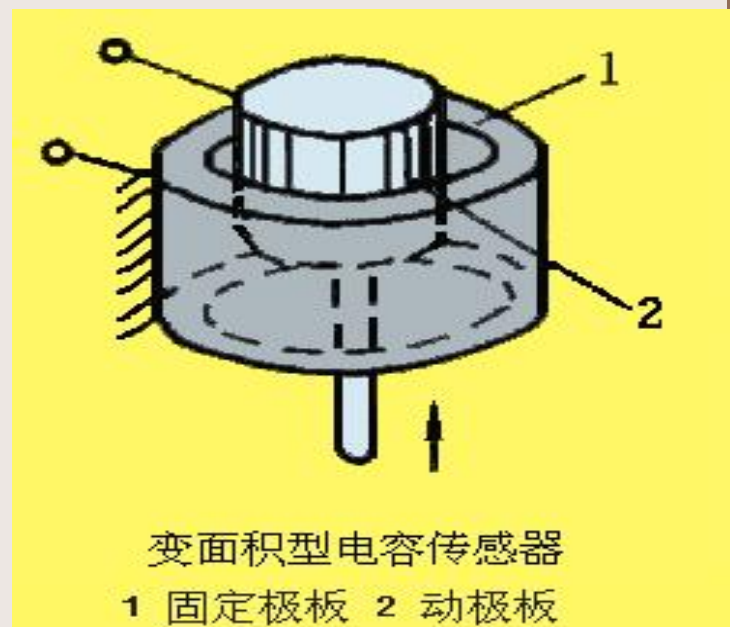


$$\Delta u_o = -\frac{u_{ref}}{Z_1(1+Z_4/Z_3)} \Delta Z$$

## 8.2.4 Coaxial Cylindrical Capacitive Sensor

圆柱体线位移型：

由于平板型传感器的动极板稍向间距方向变化时，将会影响测量精度。因此，可将变面积型电容式传感器做成同轴圆柱形极板结构。

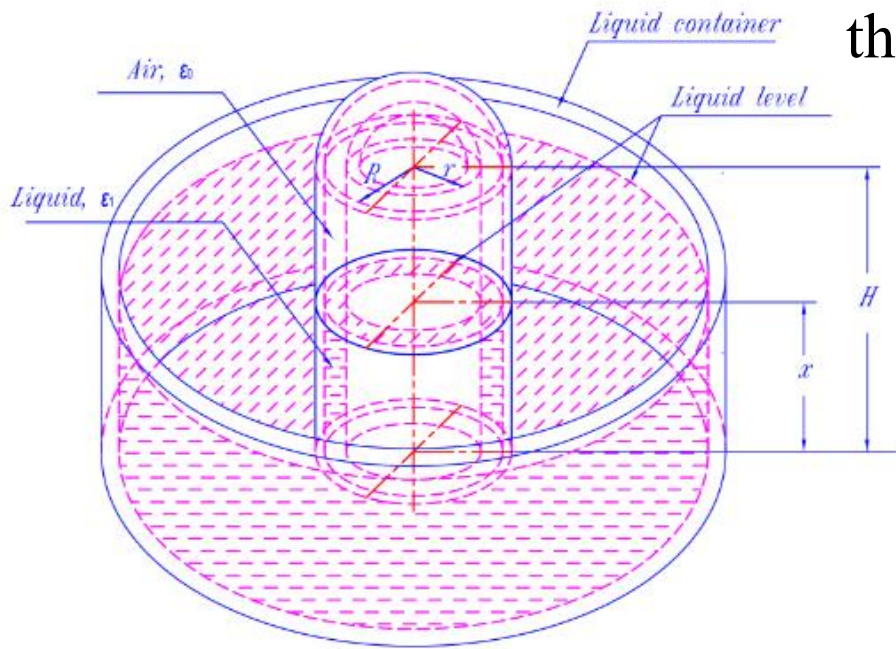


$$C = \frac{2\pi\epsilon_0\epsilon x}{\ln(D/d)} \quad S = \frac{dC}{dx} = \frac{2\pi\epsilon_0\epsilon}{\ln(D/d)} = \text{常数}$$

**特点：**输出与输入成线性关系，与极距型相比灵敏度较低，适于较大直线位移及角位移测量。

## 8.3 Variable Permittivity Capacitive Sensors

The dielectric constant variation between the two plates of the capacitive sensor will cause the change of the capacitance as the dielectric changes.



the capacitance of the gas medium

$$C_1 = \frac{2\pi\epsilon_0(H-x)}{\ln(R/r)}$$

capacitance of the liquid medium

$$C_2 = \frac{2\pi\epsilon_1 x}{\ln(R/r)}$$

## 8.3 Variable Permittivity Capacitive Sensors

total capacitance

$$\begin{aligned} C &= \frac{2\pi\varepsilon_0(H-x)}{\ln(R/r)} + \frac{2\pi\varepsilon_1 x}{\ln(R/r)} = \frac{2\pi\varepsilon_0 H}{\ln(R/r)} + \frac{2\pi(\varepsilon_1 - \varepsilon_0)x}{\ln(R/r)} \\ &= C_0 + \frac{2\pi(\varepsilon_1 - \varepsilon_0)}{\ln(R/r)} x \end{aligned}$$

$H$  is the height of the capacitor plates,  $R$  is the inner radius of the outer cylinder,  $r$  is the outer radius of the inner cylinder,  $x$  is height of the liquid level,  $C_0$  is initial capacitive value determined by the structural sizes of the liquid level meter

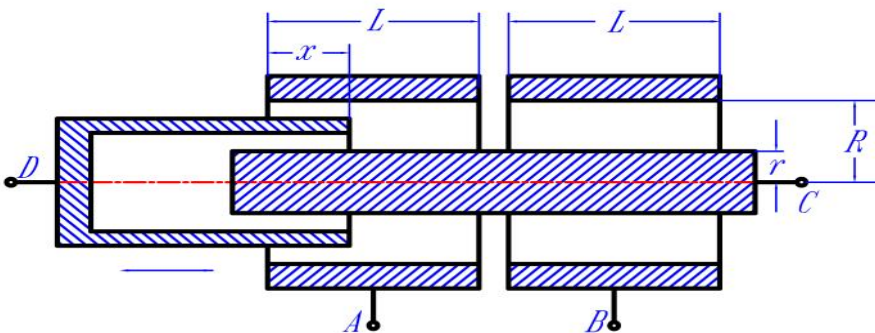
$$C_0 = \frac{2\pi\varepsilon H}{\ln(R/r)}$$



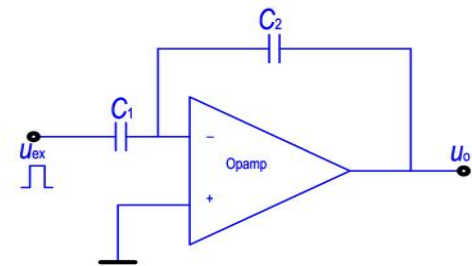
## 8.3 Variable Permittivity Capacitive Sensors

### Example

There is a capacitive displacement sensor, and its structure is shown in Figure 8.19 (a). Cylinder C is the inner electrode and Cylinder A and B are two outer electrodes, and D is a shield sleeve, CB forms a fixed capacitance  $C_F$ , and CA Forms a variable capacitance  $C_x$  which changes with the input displacement  $x$  of the movable shielding sleeve.



(a) Structural Schematic Diagram



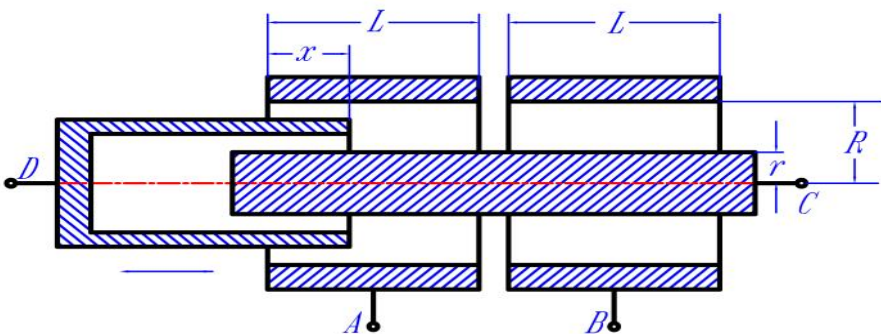
(b) Operational Amplifier Circuit

## 8.3 Variable Permittivity Capacitive Sensors

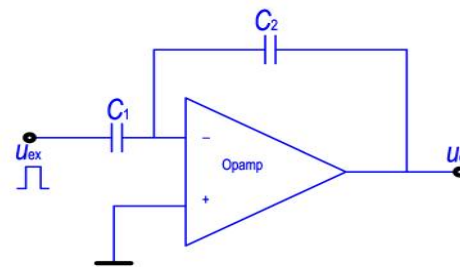
Knowing that  $l = 25 \text{ mm}$ ,  $R = 6 \text{ mm}$ ,  $r = 2 \text{ mm}$ , ideal operational amplifier circuit is adopted, as shown in Figure 8.19 (b), the effective value of the signal source voltage is  $u_{\text{ex}} = 6 \text{ V}$ .

Question:

- (1) Sensitivity of this capacitive displacement sensor.
- (2) In order to keep the operational amplifier output voltage  $u_o$  linear with the displacement  $x$ , please put the fixed capacitance  $C_F$  and the variable capacitance  $C_x$  in right position in the circuit, and calculate the sensitivity value of the output voltage.



(a) Structural Schematic Diagram



(b) Operational Amplifier Circuit



## 8.3 Variable Permittivity Capacitive Sensors

**Solution:**

(1) Since  $C_x = \frac{2\pi\epsilon_0(L-x)}{\ln(R/r)}$ ,

The sensitivity of the capacitive sensor is given by

$$S_x = \left| \frac{dC_x}{dx} \right| = \frac{2\pi\epsilon_0}{\ln(R/r)} = \frac{2 \times 3.14 \times 8.85 \times 10^{-12}}{\ln(6 \times 10^{-3} / 2 \times 10^{-3})} \approx 0.05 \times 10^{-9} \text{ (F/m)}$$

(1) Since  $u_o = -u_{ex} \frac{C_1}{C_2}$ ,  $C_x = \frac{2\pi\epsilon_0(L-x)}{\ln(R/r)}$ ,  $C_F = \frac{2\pi\epsilon_0 L}{\ln(R/r)}$ ,

To make the output voltage  $u_o$  keep linear relation with the displacement  $x$ , there are  $C_1 = C_x$ ,  $C_2 = C_F$ ,

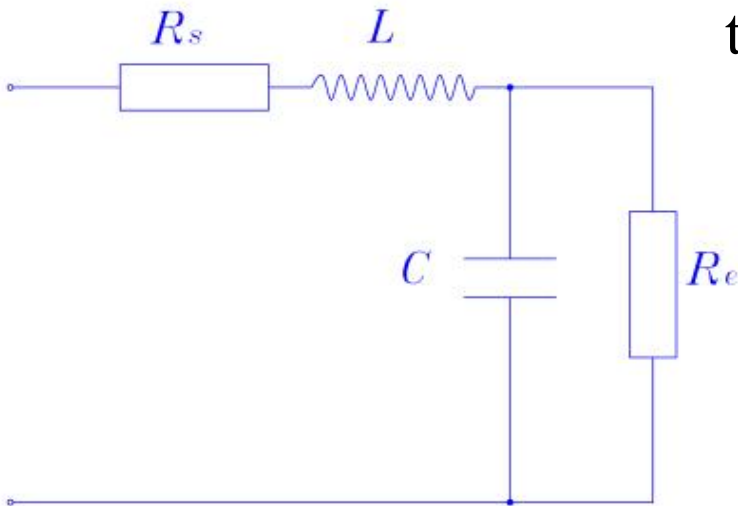
Then  $u_o = -u_{ex} \frac{C_x}{C_F} = -u_{ex} \frac{L-x}{L}$ ,

Thus, the sensitivity of the output voltage is given by

$$S_V = \left| \frac{du_o}{dx} \right| = \frac{u_{ex}}{L} = \frac{6}{25} = 0.24 \text{ (V/mm)}$$

## 8.4 Measuring Circuit of Capacitive Sensor

### 8.4.1 Equivalent Circuit



the equivalent reactance

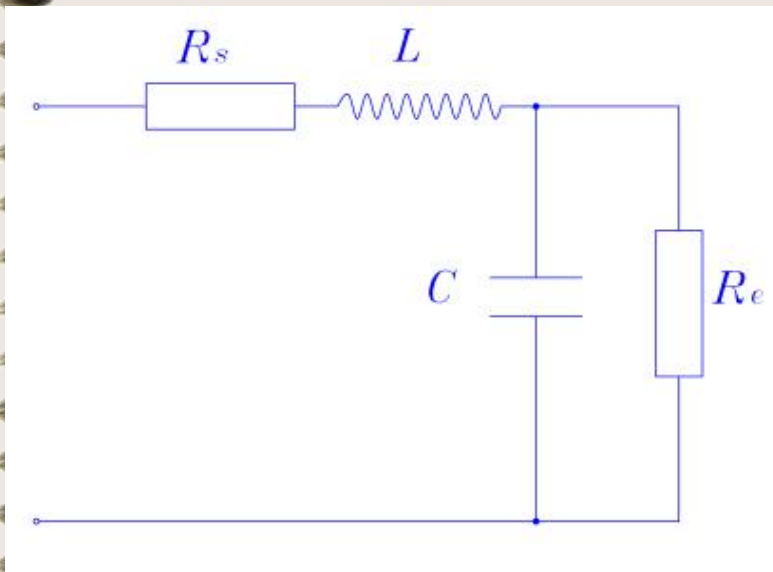
$$Z_c = \left( R_s + \frac{R_e}{1 + \omega^2 R_e^2 C^2} \right) - j \left( \frac{\omega R_e^2 C}{1 + \omega^2 R_e^2 C^2} - \omega L \right)$$

Because the parallel resistance  $R_e$  of the sensor is very large and the series resistance  $R_s$  is very small

$$Z_c = \frac{1}{j\omega C_e} = \frac{1}{j\omega C} + j\omega L$$

## 8.4 Measuring Circuit of Capacitive Sensor

### 8.4.1 Equivalent Circuit



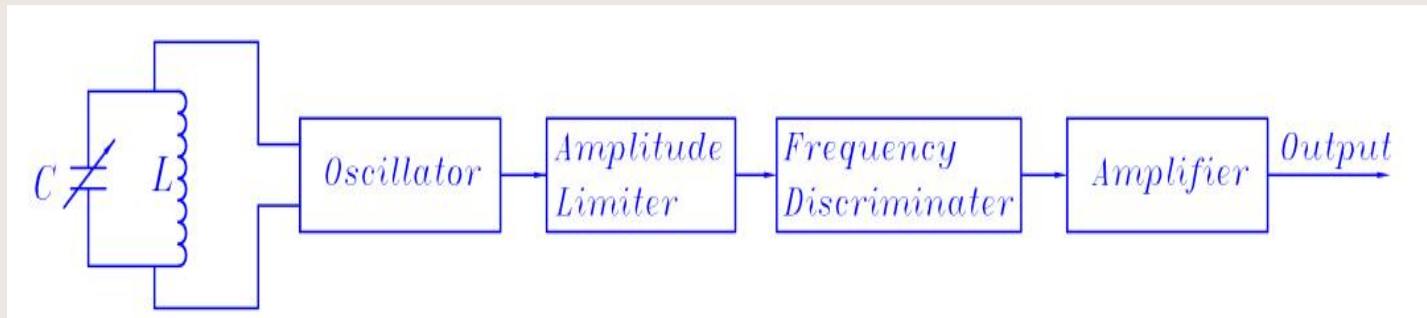
the equivalent capacitance of the sensor is written as

$$C_e = \frac{C}{1 - \omega^2 LC} = \frac{C}{1 - (f/f_0)^2}$$

Where  $f_0 = 1/(2\pi\sqrt{LC})$  , is resonant frequency of the equivalent circuit

## 8.4 Measuring Circuit of Capacitive Sensor

### 8.4.2 Frequency Modulation Circuit



**电容→频率→电压**

传感器电容是振荡器谐振回路的一部分。

传感器电容的变化→振荡频率变化→鉴频器  
将频率变为电压。

## 8.5 Characteristics and Application

### 8.5.1 Characteristics of Capacitive Sensors

- (1) Simple structure, good temperature stability and strong adaptability
- (2) High sensitivity, fast dynamic response and wide range
- (3) Non-contact measurement and no influence on the measured object
- (4) High output impedance, low power output and poor load capacity
- (5) Significant influence from parasitic capacitance and poor anti-interference ability
- (6) Edge effect and nonlinear output characteristic

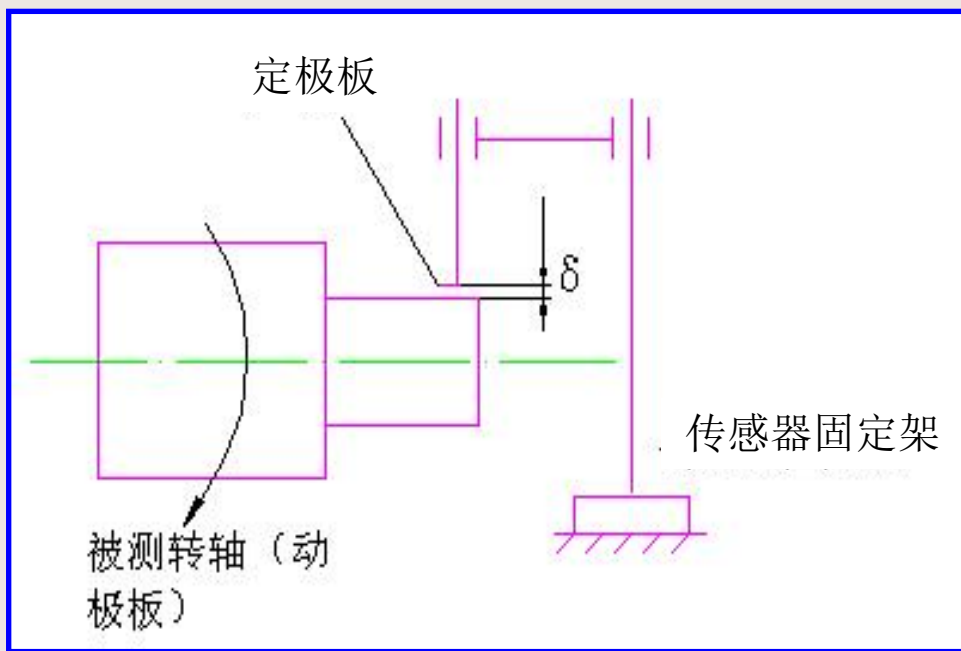


## 8.5 Characteristics and Application

### 8.5.2 Application

- 利用极板间距和极板覆盖面积 $A$ 的变化，可以测量 **直线位移或角位移**；
- 通过弹性元件也可以 **测量力、压力、振动或加速度**等；
- 利用介电常数的变化还可以进行一些特殊量的测量，**如液位、浓度、厚度、湿度**等。

## 应用：测量径向跳动

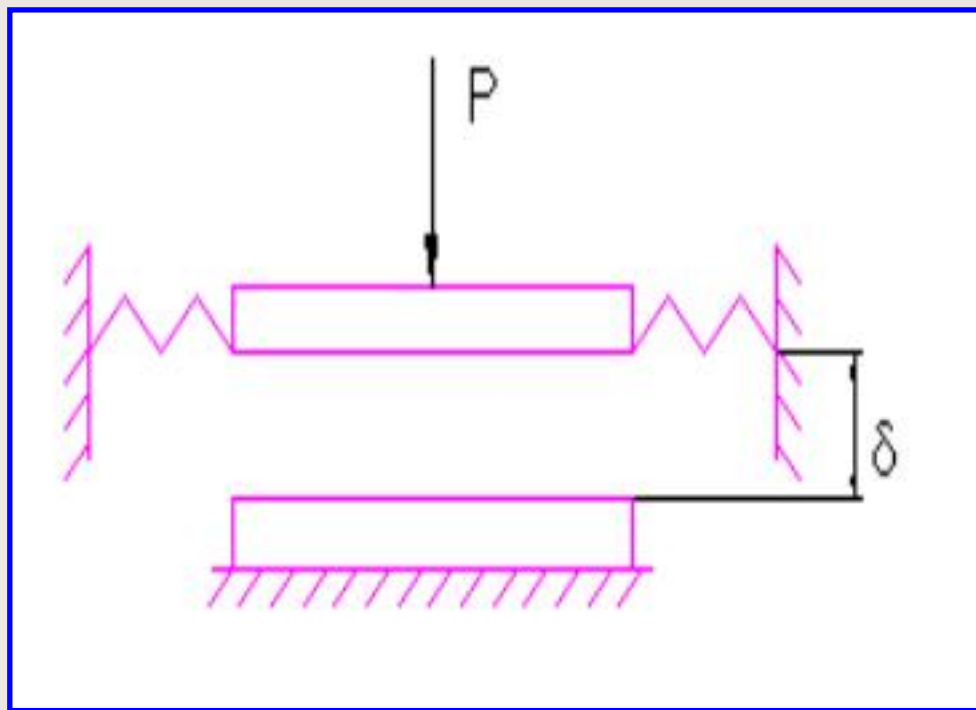


测轴的径向跳动  
(小位移)

轴旋转时的径向跳动 ( $\delta$ 变化)  $\rightarrow$   $C$ 变化  $\rightarrow$  可测轴的跳动 (小位移)。

## 应用：测量力

力 $P$ 使 $\delta$ 变化 $\rightarrow C$ 变化，测 $C$ 可知 $P$ 力大小



# 应用：传声 Condenser Microphone

$$C = \frac{\epsilon_0 \epsilon A}{\delta}$$

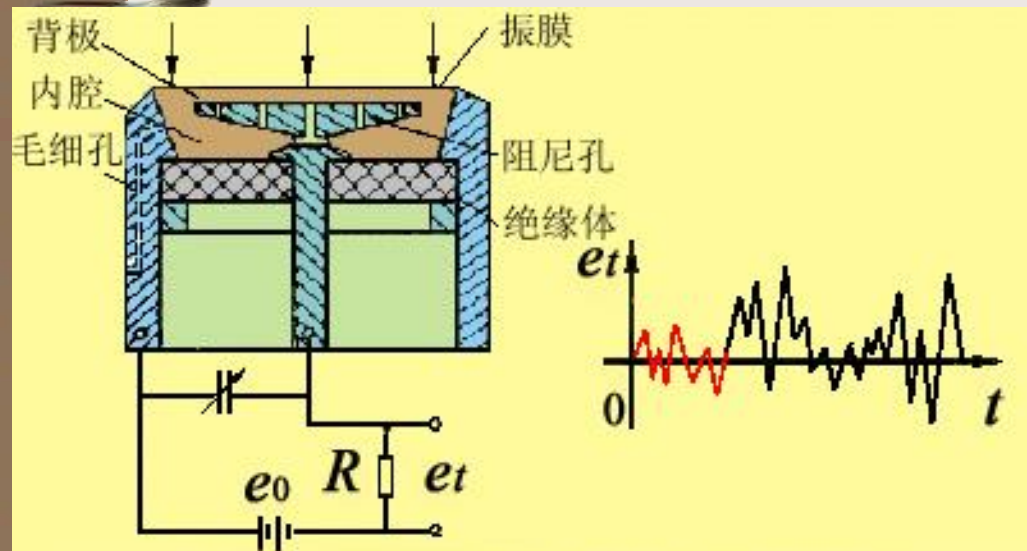
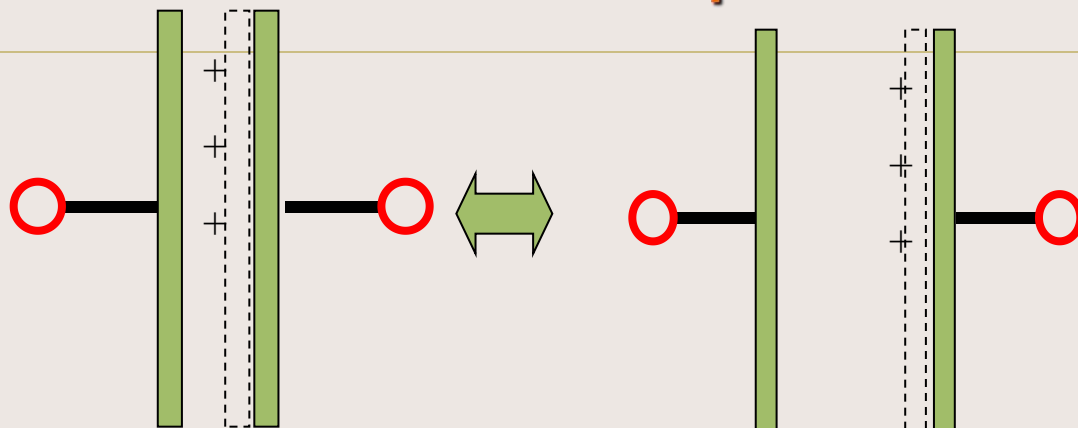
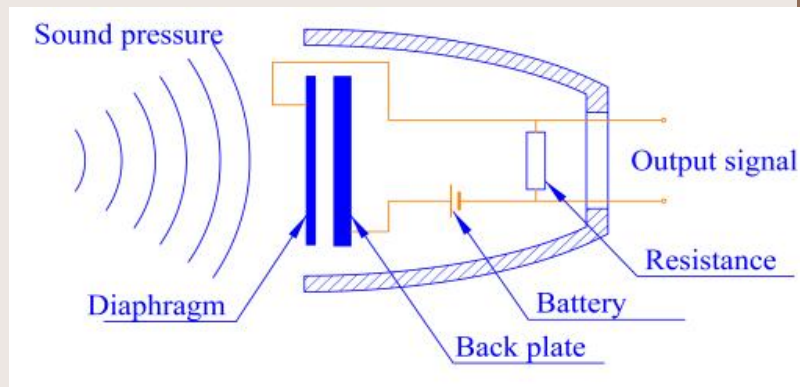
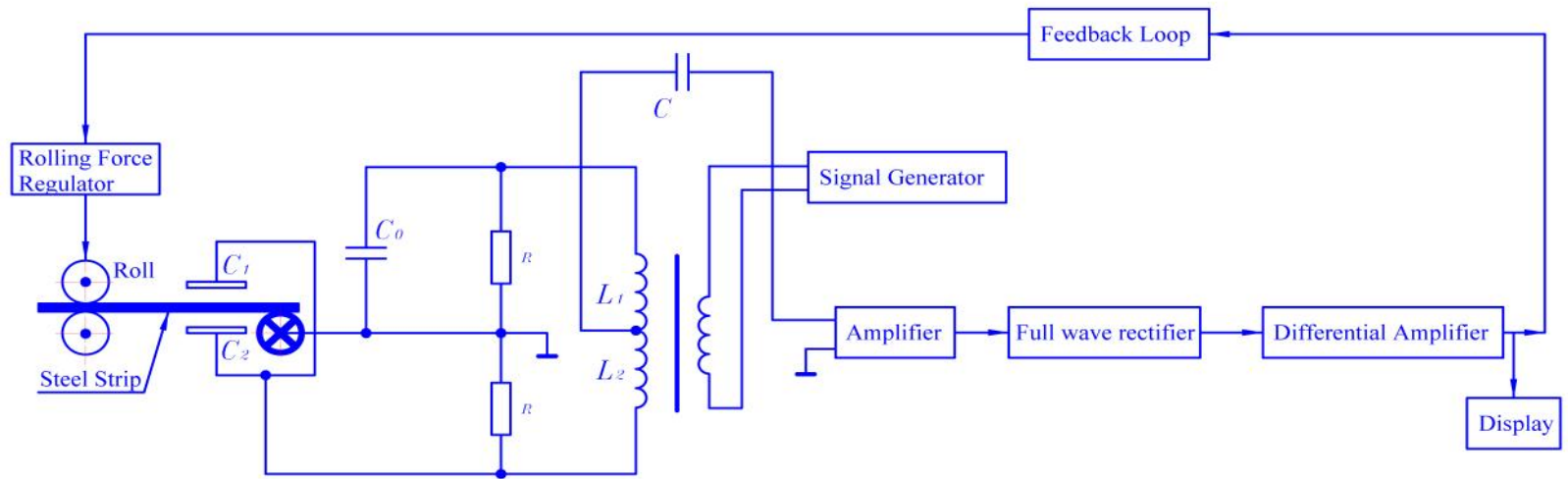


图8.5-5b 电容式传声器的电路原理



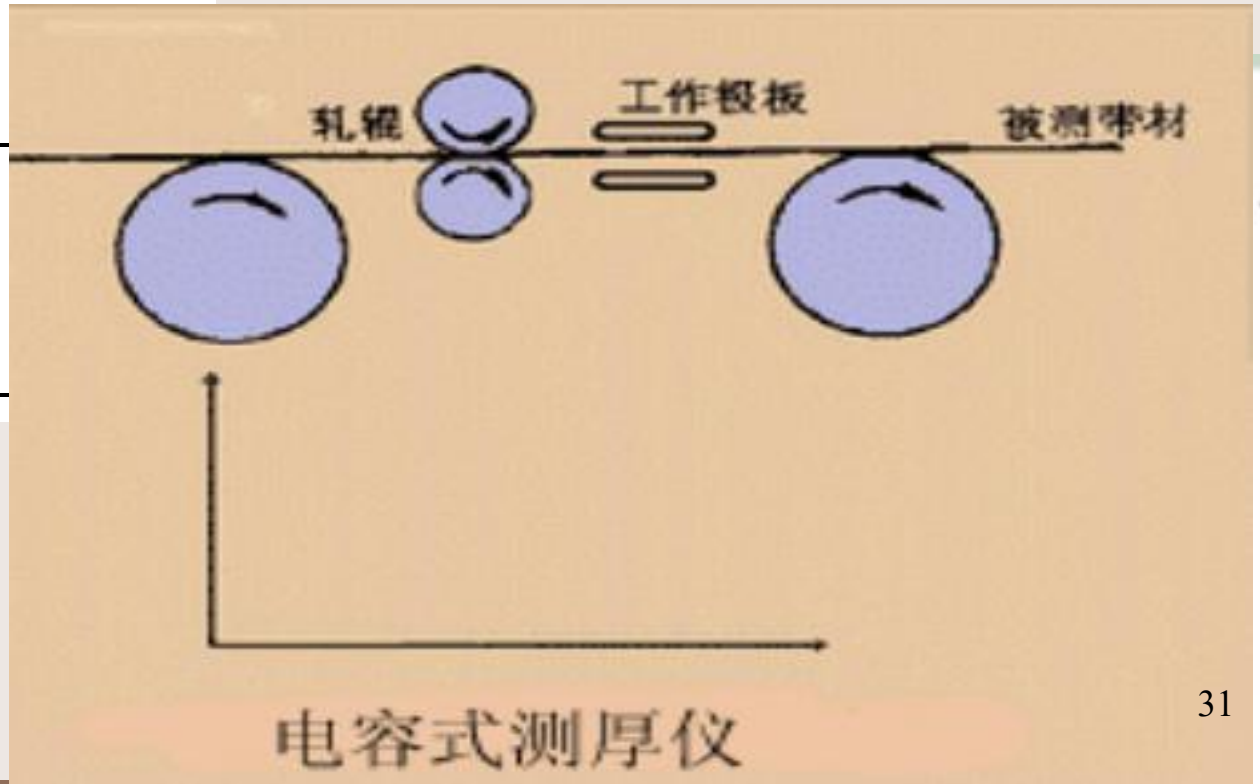
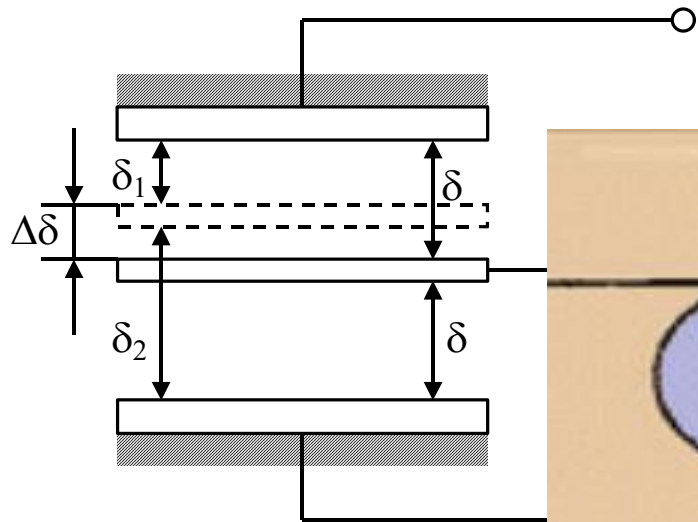
# Thickness Measuring



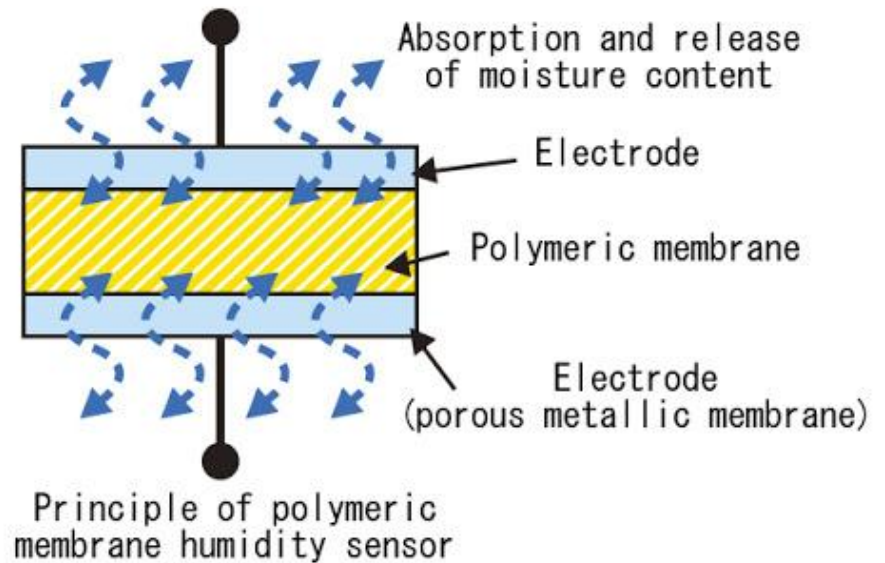
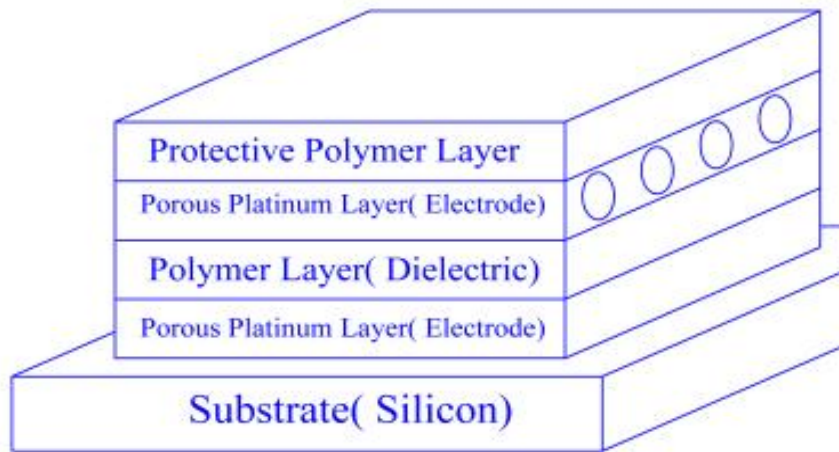
$$\begin{aligned}
 \Delta u &= \Delta u_1 + \Delta u_2 = u_o \frac{C_0}{\epsilon A} \Delta h \\
 &= u_o \frac{C_0}{\epsilon A} (\Delta h_a + \Delta h_b)
 \end{aligned}$$



# Thickness Measuring



## Capacitive Humidity Sensor



As the moisture in the atmosphere changes its permittivity, a simple Capacitive Relative Humidity Sensor can be made from an air-filled capacitor.



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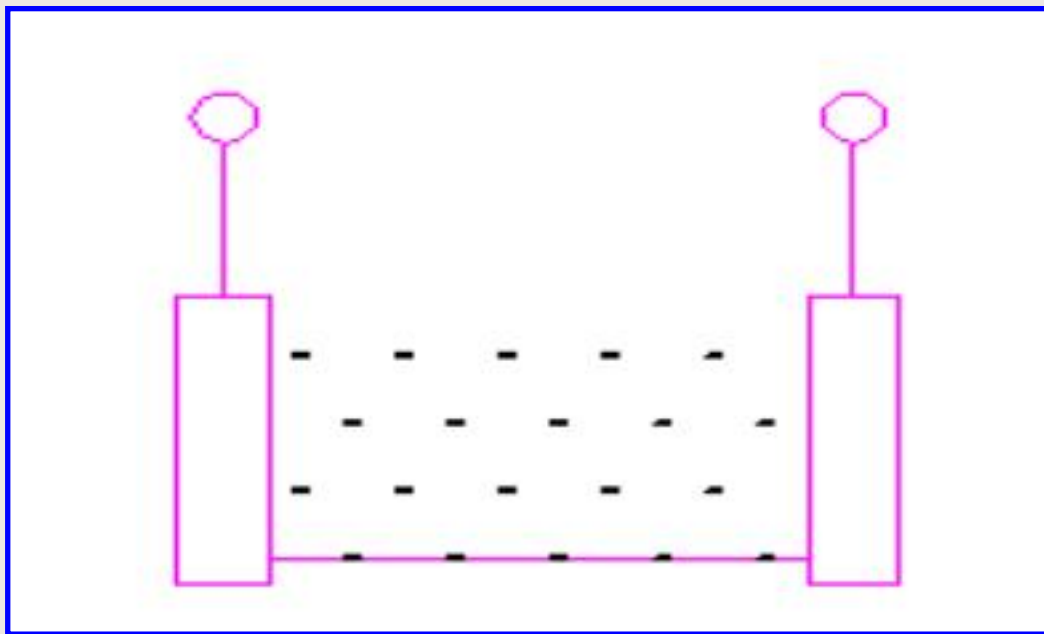
# 电容式传感器

产品.

液  
作  
瓷  
表  
膜  
位



**应用:** 液位高度、材料的厚度与介电常数、湿度，以及物料的水份。

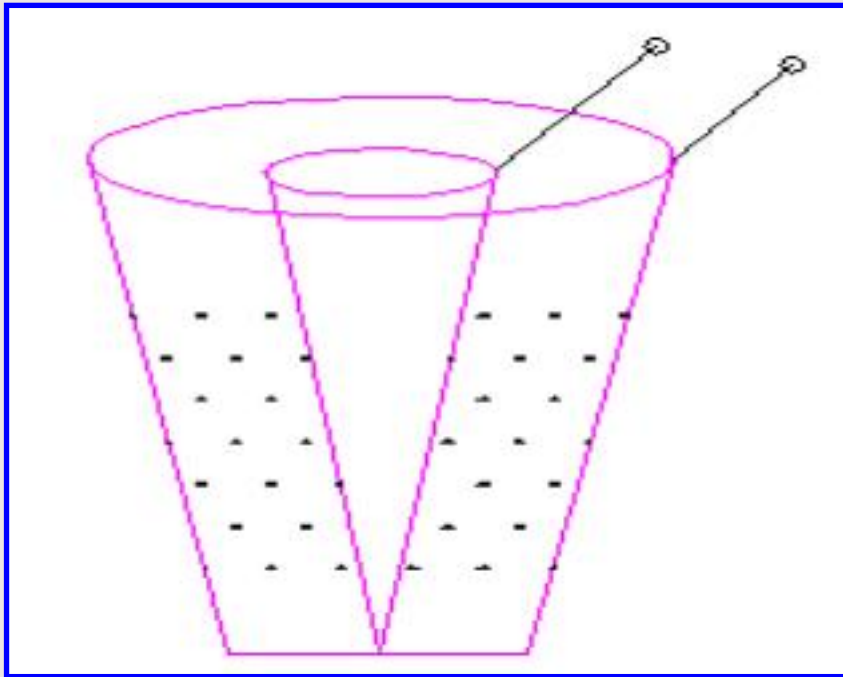


测谷物土壤水份，空气 $\epsilon=1$

水 $\epsilon=81$

完全干燥谷物 $\epsilon=2$

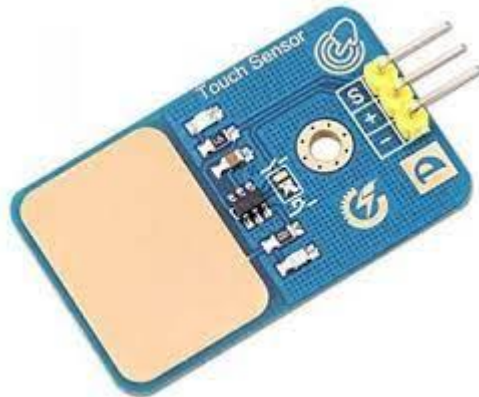
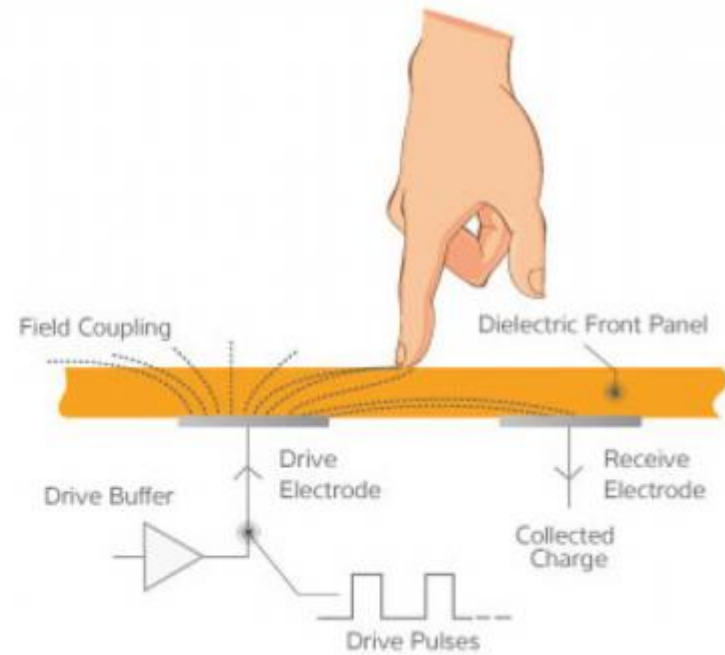
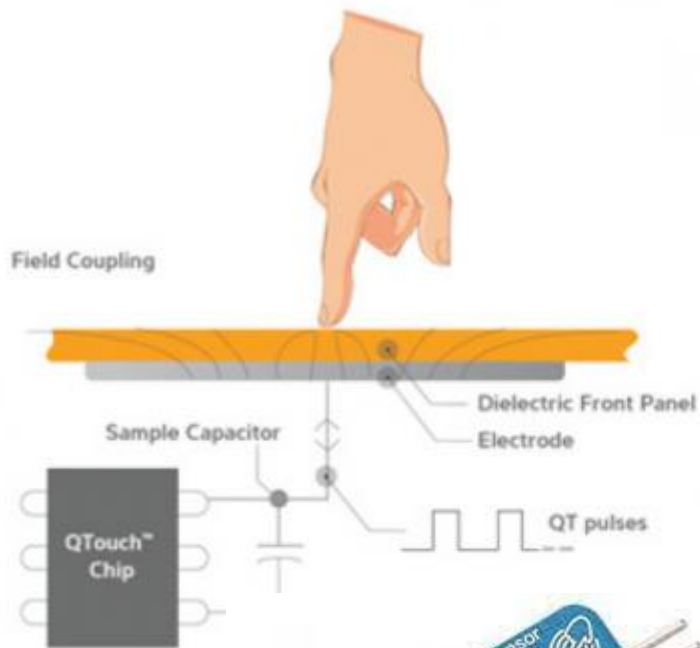




插入式谷物  
水份测量仪

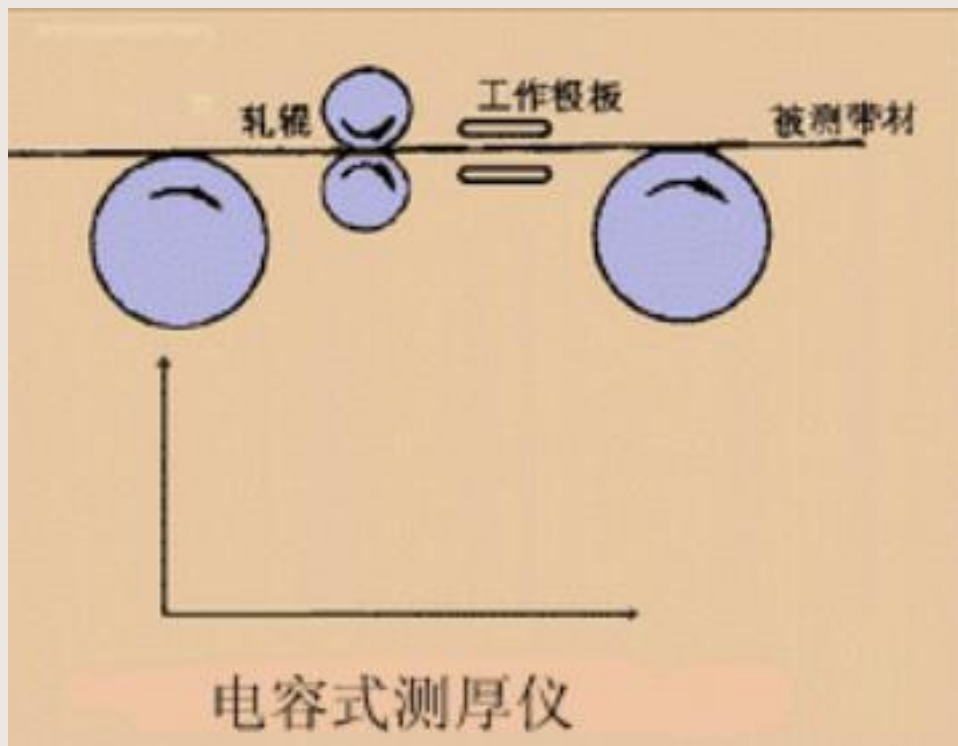


# Touch sensor



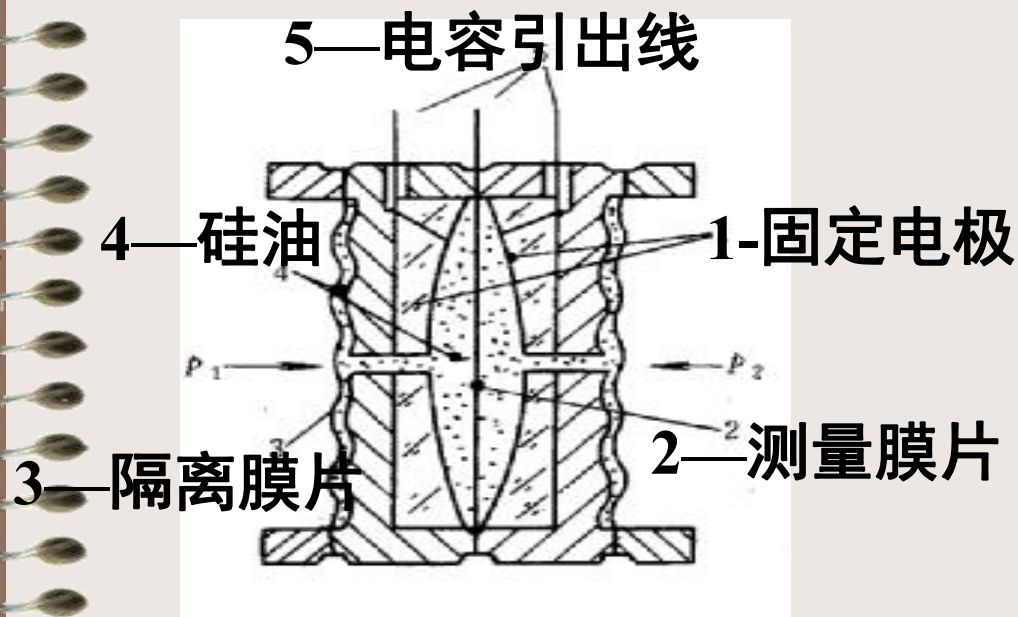
## 电容测厚仪

下图为测量厚度的电容测厚仪原理图。总电容 $C_x = C_1 + C_2 = 2C$ 。如果带材厚度发生变化，则引起电容量的变化。



## 电容式压力传感器

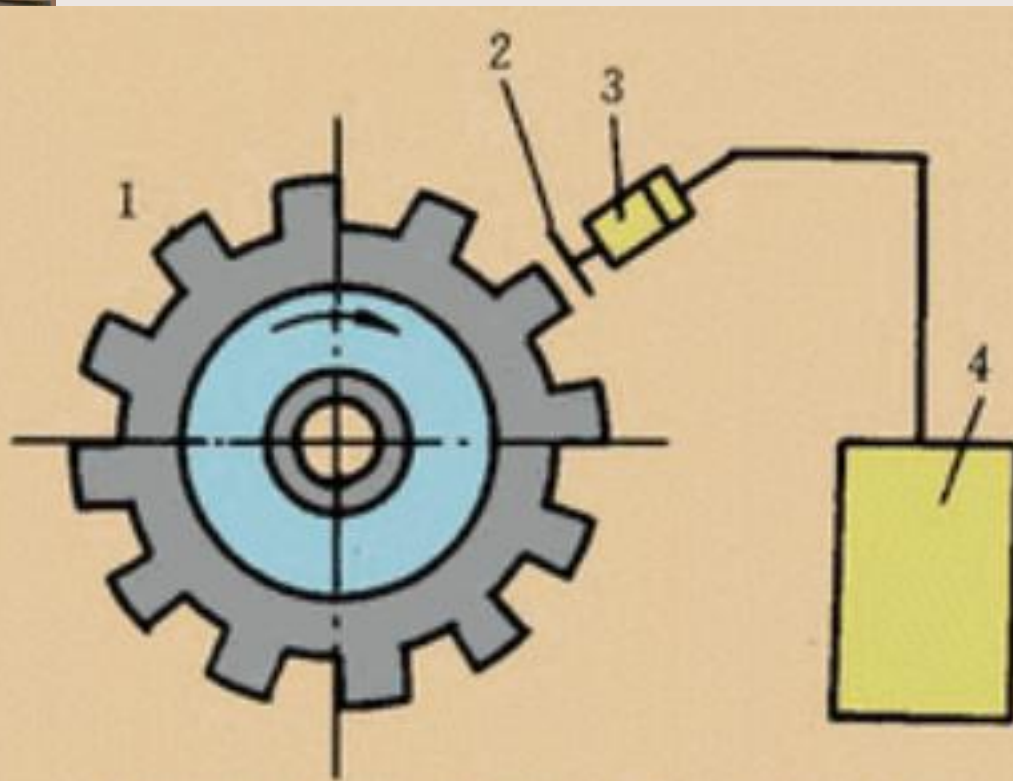
电容式压力传感器一般都利用弹性膜片作为敏感元件，该传感器主要由一个动电极、两个固定电极和这三个电极的引出线组成。动电极为圆形薄金属膜片，它既是动电极，又是压力的敏感元件，固定电极为中凹的镀金玻璃圆片。



## 陶瓷电容压力传感器

液体压力  
作用在陶  
瓷膜片的  
表面，使  
膜片产生  
位移。





电容式转速传感器的结构原理

1- 齿轮 2- 定级

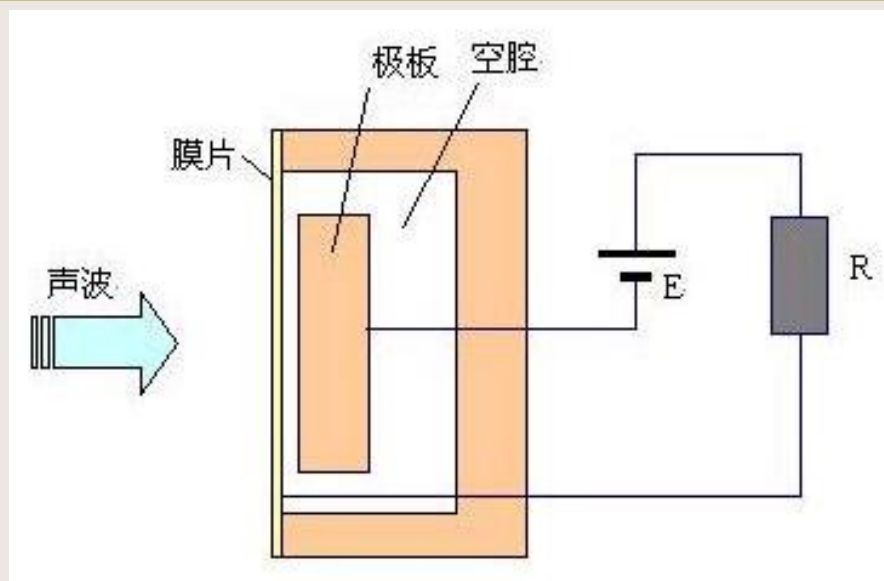
3- 电容式传感器 4- 频率计

齿轮转动时，电容量发生周期性变化，通过测量电路转换为脉冲信号，则频率计显示的频率代表转速大小。设齿数为 $z$ ，频率为 $f$ ，则转速为：

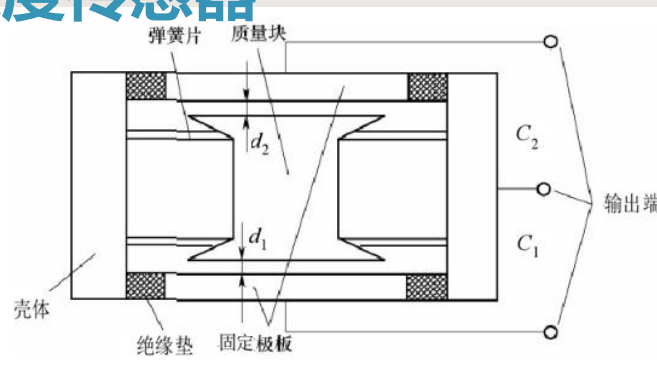
$$n = \frac{60f}{z} \quad (r/min)$$



## ■ 电容式传声器



## ■ 电容式加速度传感器



# 电容式液位计

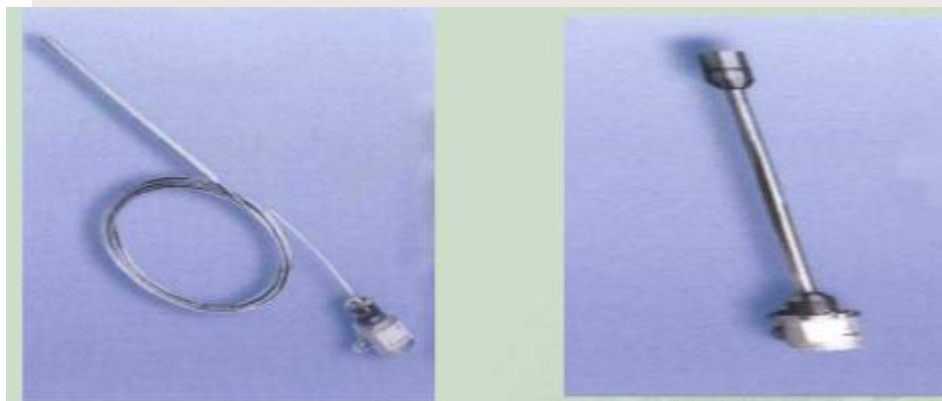
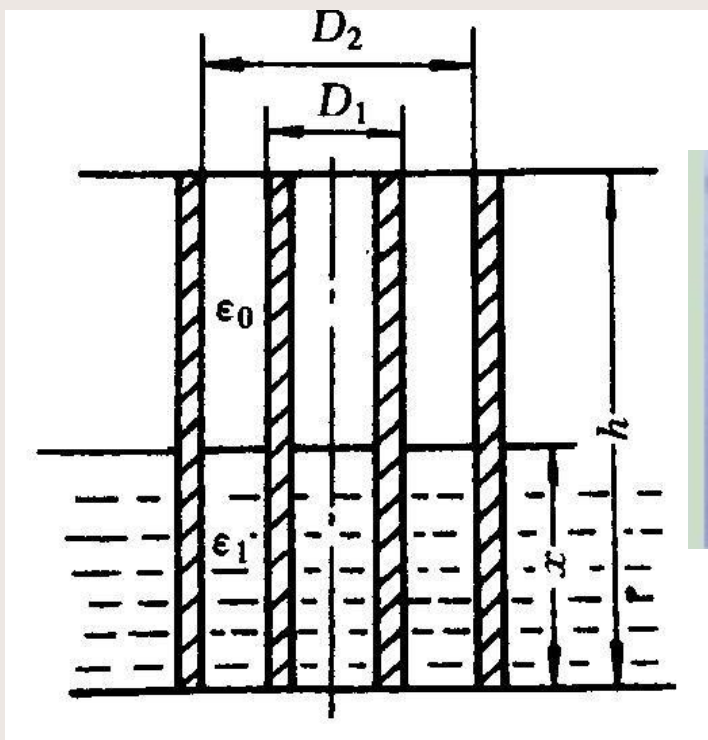
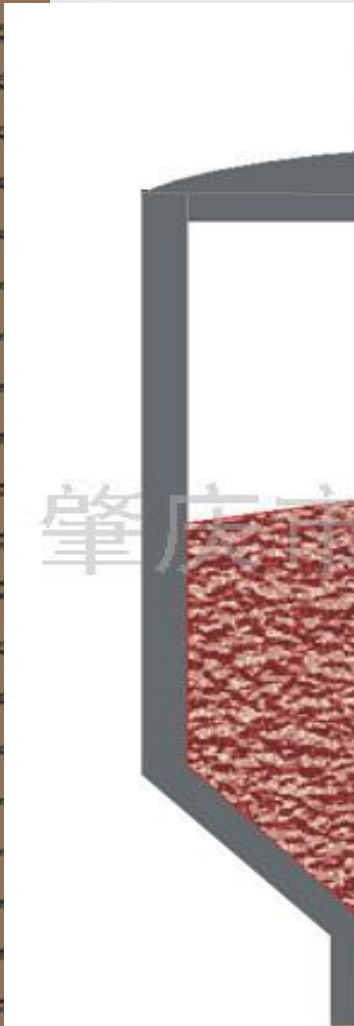


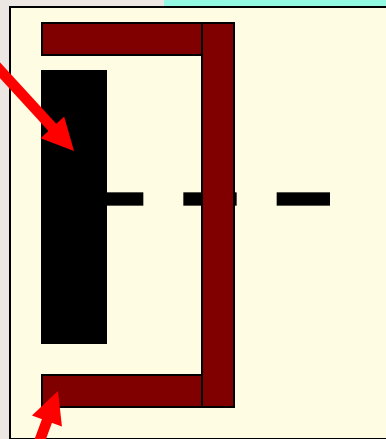
图7.3.3 电容式液位计原理图

# 电容式液位传感器

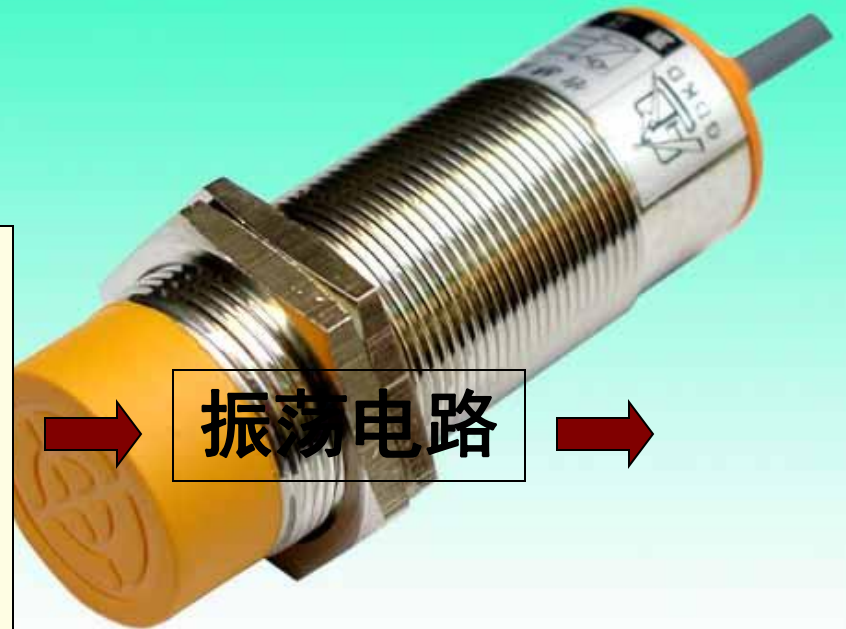


## 电容式接近开关

被测物体 感应电极



被测电容



振荡电路



谢谢！

